

AI 507: Artificial Intelligence and Society



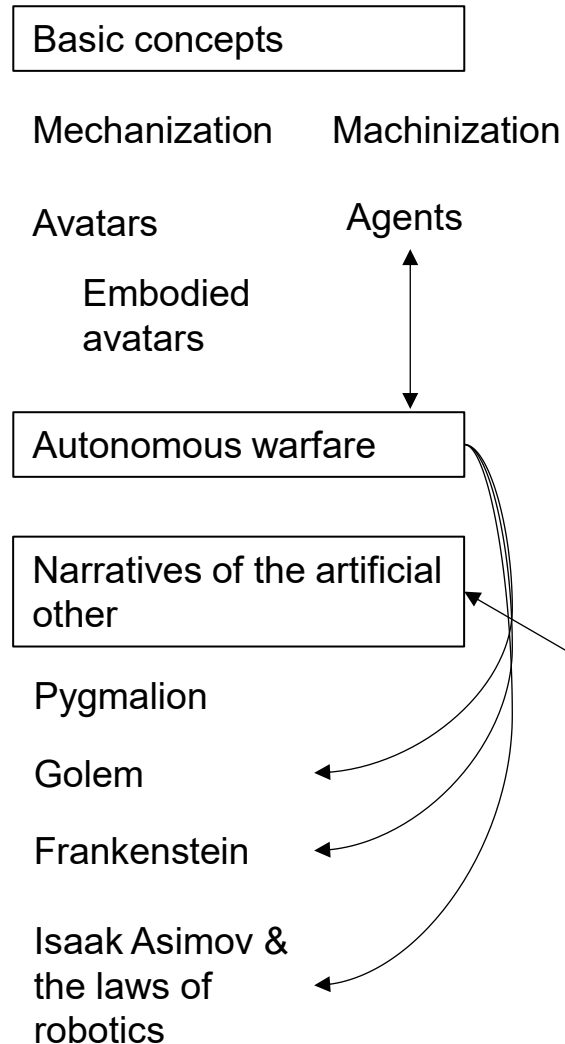
#7 Algorithms

Associate Prof. Dr. Lena Frischlich

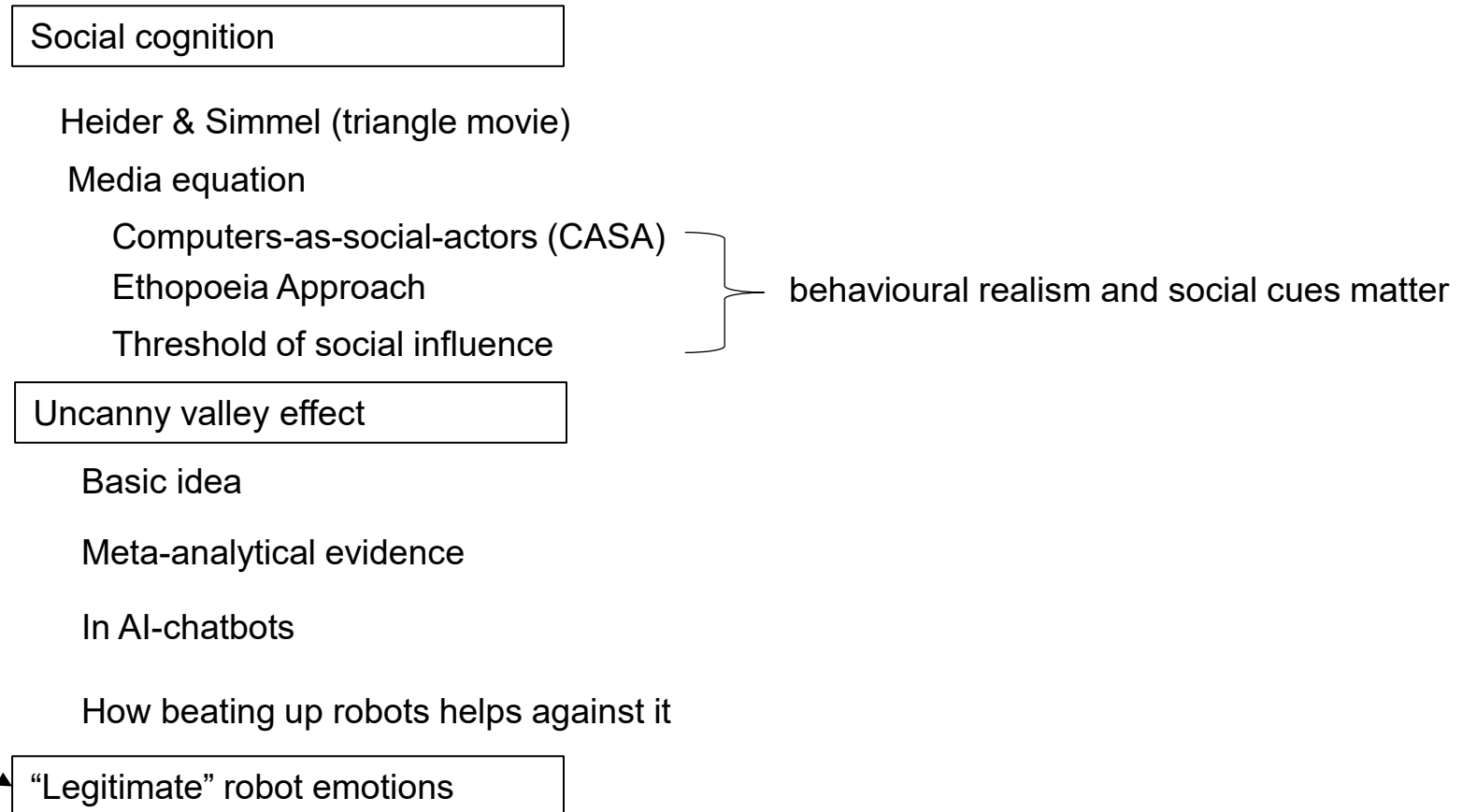
Flashlight – Video session

Flashlight

Robots



Perception of artificial others



Any questions on that?



Then let's get started!

After today's lecture, you will hopefully...

- ☐ ...have a better understanding of algorithmic recommendation
- ☐ ...have a better idea about algorithmic literacy (or the lack thereof)
- ☐ ...have an initial idea of how democratic values can be embedded in recommender systems
- ☐ ...understand how and why humans select content – and of microtargeting

Algorithms and algorithmic recommendation

Some of the mothers of algorithmisation

Ada Lovelace (Lady Byron),

- Worked with Charles Babbage at the British high court
- Was the first to realise that Babbage's Analytical Engine could calculate almost anything that could be translated into numbers
- Ada wrote a complicated calculation as an example for Babbages paper (its in the appendix) → considered the first algorithm for a computer (which was never built)



Bleechly Girls such as **Joan Clarke** who worked as codebreakers and with Turing on the Enigma machine for deciphering the codes of the Nazis

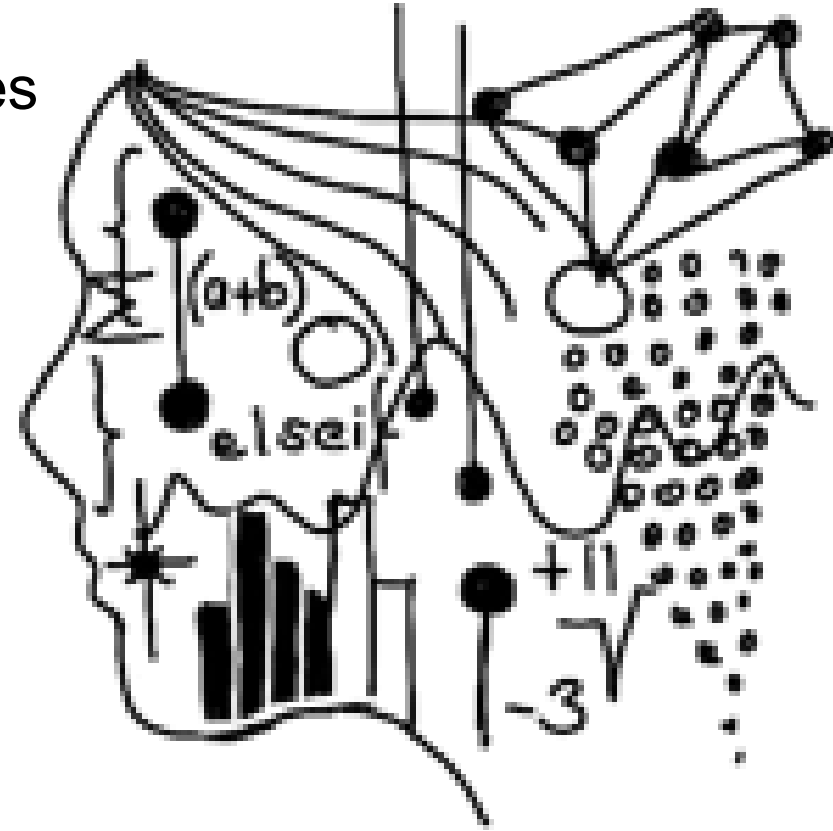
The first general-purpose digital computer, ENIAC, was developed in 1945 – mostly by

Jean Jennings, Marlyn Wescoff, Ruth Lichtermann, Betty Hoberton Frances Spence and Kathleen Antonelli



Today: algorithms are everywhere (Chollet & Allaire, 2018)

- All processes that take place in computers or smartphones are ultimately based on algorithms
- They are at the heart of almost all technological advancements
- Centrally – of course – in artificial intelligence (AI)
- In most digital environments content is selected and presented based on algorithmic recommendation
- But how does it work?



“Traditional” approaches to recommendation

(Casalegno, 2022; Da’u, & Salim, 2020;)

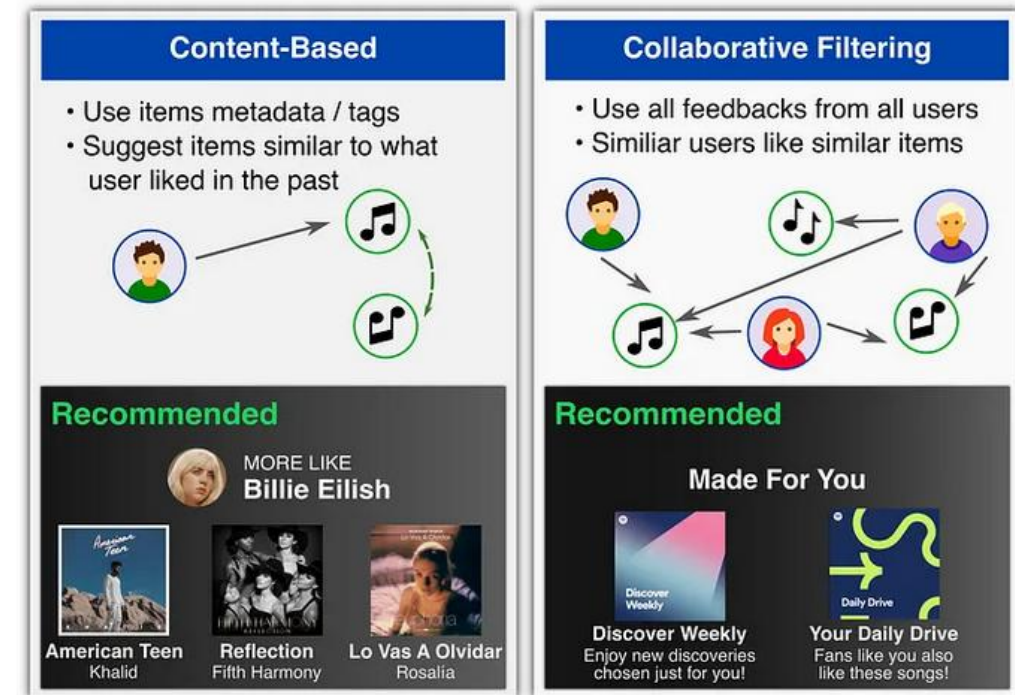
Collaborative Filtering:

- The behavior of users and their social contacts influences what is recommended
- Needs information about past behavior & the social graph
- Very frequent type of recommenders
- But: Cold start problem, data sparsity, privacy intrusion

Content-based filtering:

- Identification of patterns in the data (e.g., topics, genres, keywords)
- But: “obvious” recommendations/ homogeneity, better for new items than for new users

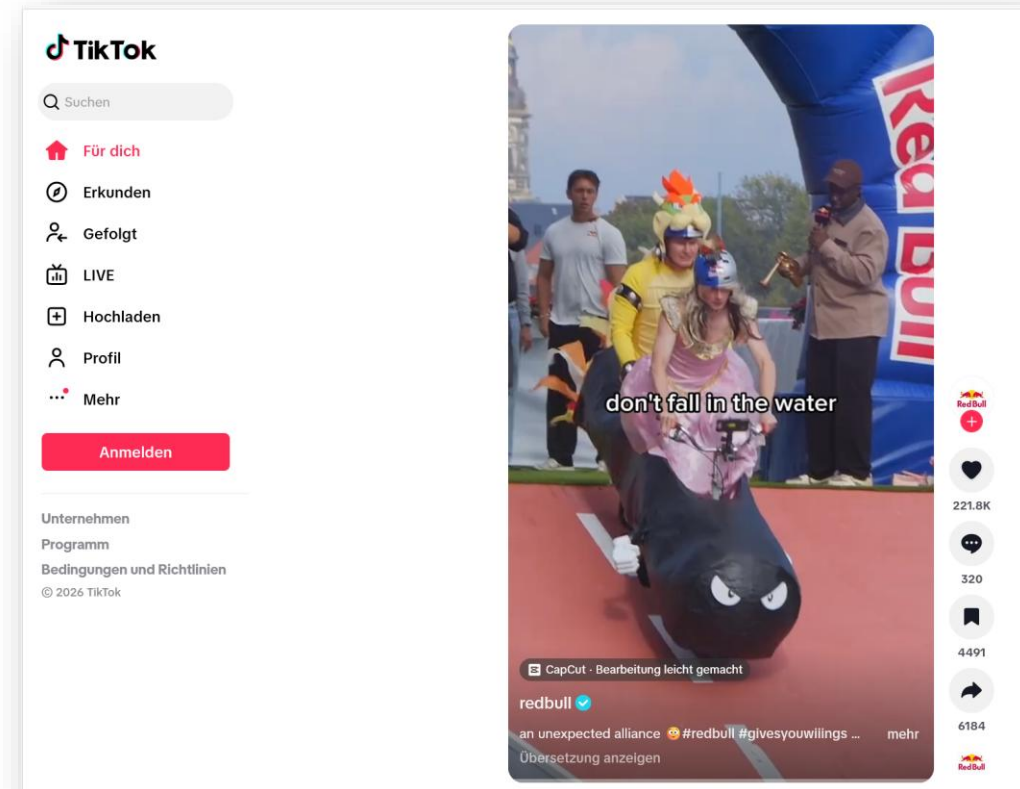
Hybrid approaches



Deep learning approaches

(Da'u & Usman Katsina, 2020; Malitesta et al., 2025)

- In the last decade, deep learning approaches became more relevant for recommendation systems
- Often: reliance on RNNs (Recurrent neural networks) and auto-encoders
- Increasingly relevant: Multi-modal recommender systems but
 - Most research works on text or images
 - Specifically, audio is underrepresented
 - All modalities can be relevant for video content
- Requires multi-step processing (each modality + joining) → Still very challenging!



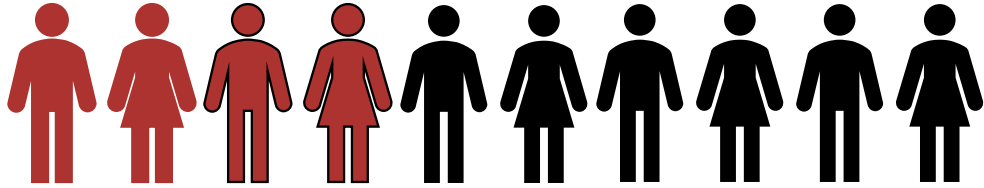
What are your questions?



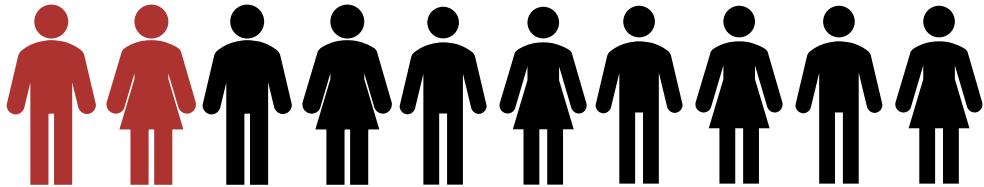
**How do people perceive
algorithmic
recommendation?**

People often trust algorithms

(Lorenzen., 2024; Smith, 2018; You et al., 2022)



40% of US-Americans agree that “algorithms can make decisions without human bias”



16-26% of Danes hold mostly negative attitudes towards AI

14-24% primarily positive ones – the majority is in the middle

- Experimental studies show that people trust algorithmic recommendations more than human ones
- Even when they learn that the algorithm makes errors

Algorithmic literacy

(DeVito, 2021; Dogruel et al., 2022; Oeldorf-Hirsch & Neubaum, 2025)

= the capacity and opportunity to...

1. be aware of the presence of algorithmically driven systems in online applications, platforms, and services
2. knowing how these systems work
3. being able to critically evaluate algorithmic decision-making
4. knowing the impact of algorithms on self- or collaboratively-identified goals
5. having the skills to cope with algorithmic operations
6. having the capacity and opportunity to crystalize this understanding into a strategic use to archive these goals

Oeldorf-Hirsch and Neubaum

685

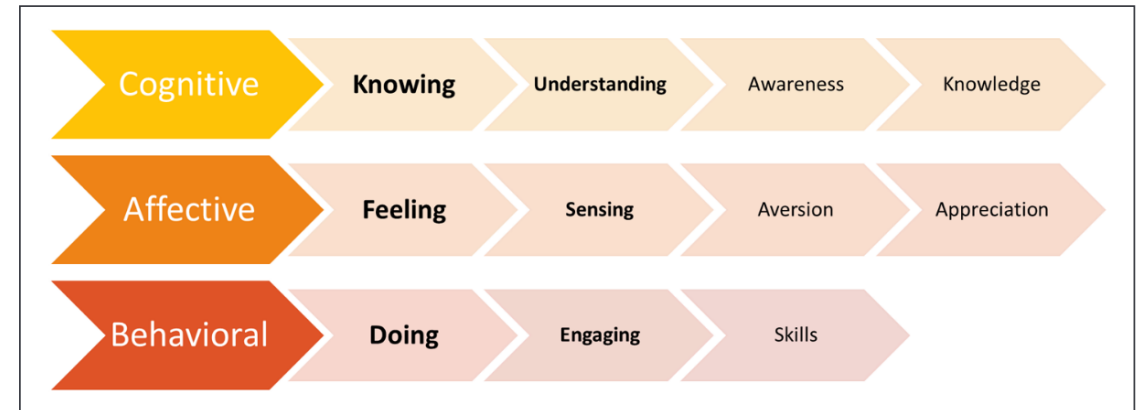


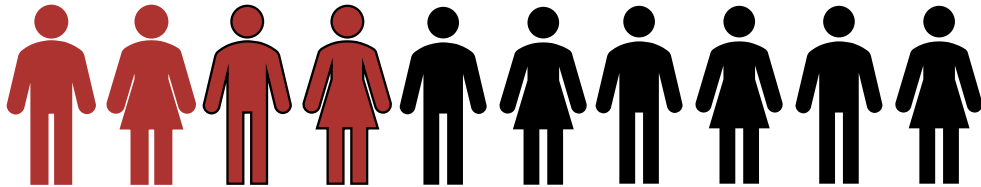
Figure 1. Dimensions of algorithmic literacy.

So: How's the state in Denmark?

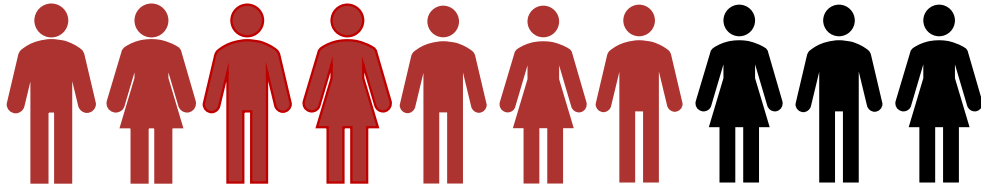
(Lorenzen et al., 2023; 2024)

7.-24. juni 2022, $N = 1337$, 18-92

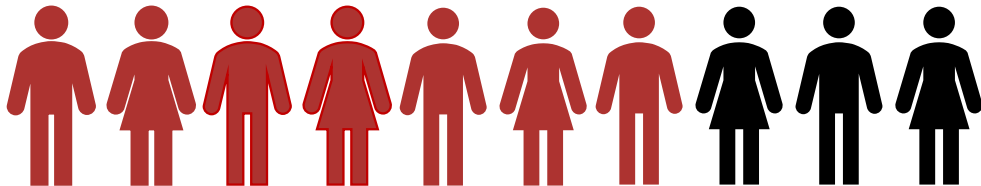
- 89% did know that Netflix' recommendations are not random
- But: only 70% knew that their google search results depend on their location



6 in 10 limit access to their personal data online



3 in 10 read the personal data policy



3 in 10 say that they could explain the difference between algorithms, machine learning and AI



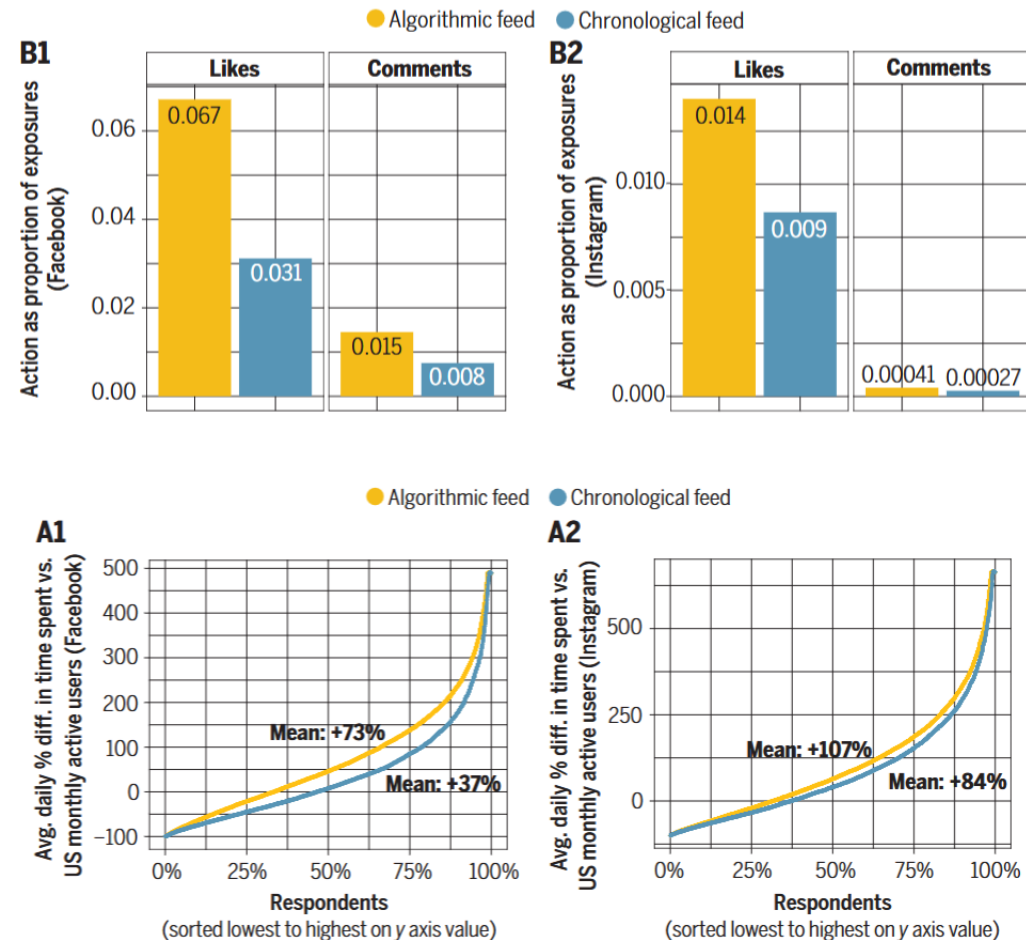
**How can algorithms
promote democracy?**

Do people want algorithms at all?

(Guess et al., 2023; Starke et al., 2025)

- Pre-registered survey, six countries (BRA; GER; JAP; SKO; UK; USA)
 $N = 6.217$
- 38% would prefer content filtering, 18% collaborative filtering, 19% random suggestions
- Approximately 20% of users would opt out of personalized recommendations (specifically in Germany)

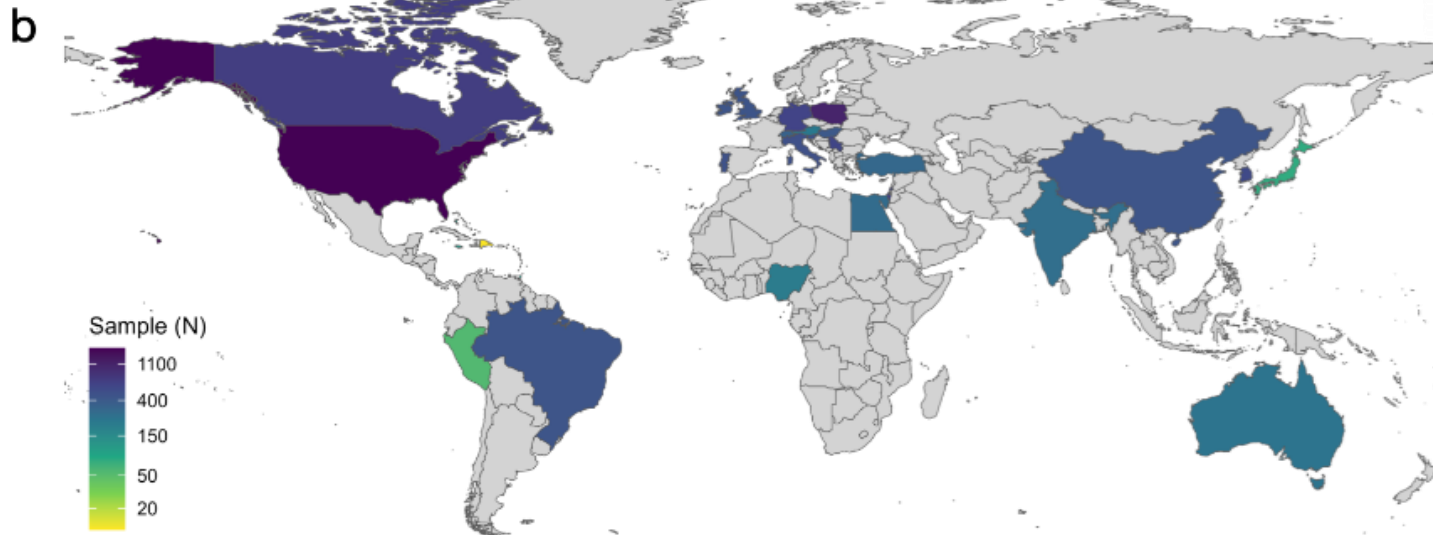
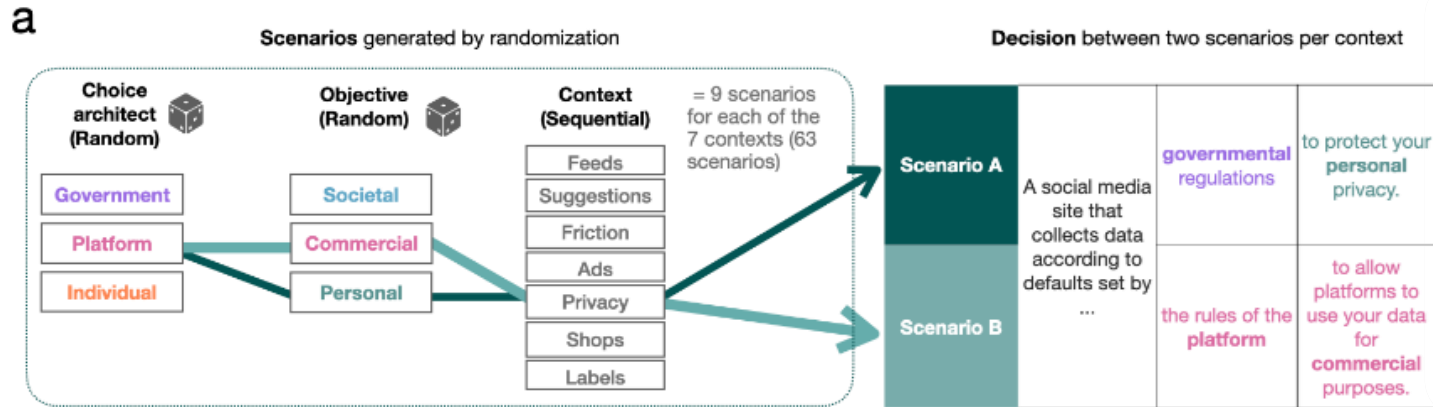
- But: Study together with Meta tested chronological feeds → less engagement



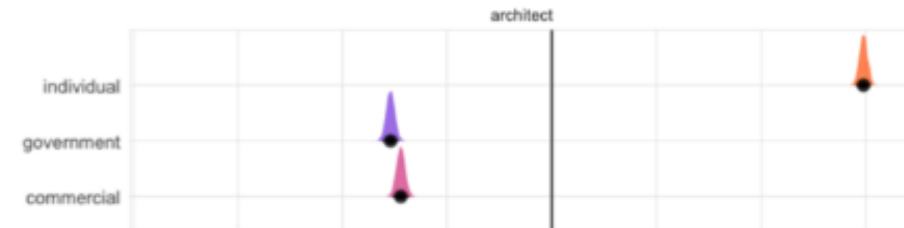
What kind of algorithms do people want?

(Stock et al., 2026)

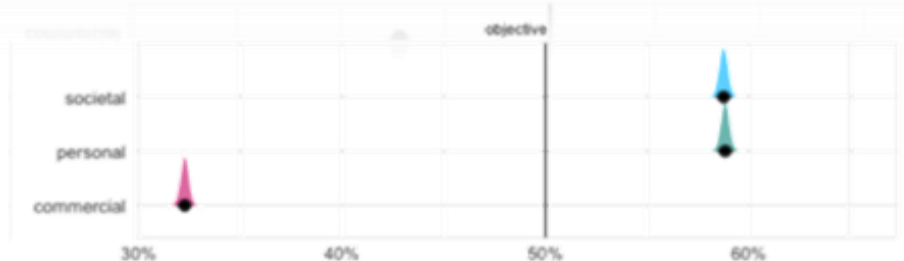
Conjoint experiment in 26 countries, $N = 11.686$



a Marginal means of probabilities for choice architect



b Marginal means of probabilities for objective



How can algorithms „for society“ look like?

(Stray et al., 2024)

Building recommender systems around human values – that is systems that are...

- **useful** (fostering control, agency, and autonomy)
- **promote well-being** (e.g. benefiting content that supports connection, community, belonging, empathy and entertainment)
- **align with human-rights and the law** (preserve privacy and freedom of expression, fair, transparent, accurate and accountable)
- **contribute to public discourse** (promoting accurate, diverse, tolerant and safe content)
- **safe**
- **align with societal values** (progress, labor, tradition, environment, duties)

What are your questions?



Time for a break



Microtargeting



Hannah Decker, postdoc DDC

Then I hope you now...

- ☐ ...have a better understanding of algorithmic recommendation
- ☐ ...have a better idea about algorithmic literacy (or the lack thereof)
- ☐ ...have an initial idea of how democratic values can be embedded in recommender systems
- ☐ ...understand how and why humans select content – and of microtargeting

AI 507: Artificial Intelligence and Society



Have a great rest of the day!

References

- Chollet, F., & Allaire, J. J. (2018). Deep Learning mit R und Keras: Das Praxis-Handbuch von den Entwicklern von Keras und RStudio. MITP-Verlags GmbH & Co. KG.
- Casalegno, F. (2022, December 12). Recommender systems—A complete guide to machine learning models. TDS Archive. <https://medium.com/data-science/recommender-systems-a-complete-guide-to-machine-learning-models-96d3f94ea748>
- Da'u, A., & Salim, N. (2020). Recommendation system based on deep learning methods: A systematic review and new directions. *Artificial Intelligence Review*, 53(4), 2709–2748. <https://doi.org/10.1007/s10462-019-09744-1>
- Devito, M. A. (2021). Adaptive folk theorization as a path to algorithmic literacy on changing platforms. *CSCW2*, 5. <https://doi.org/10.1145/3476080>
- Dogrueel, L., Masur, P., & Joeckel, S. (2022). Development and validation of an algorithm literacy scale for internet users. *Communication Methods and Measures*, 16(2), 115–133. <https://doi.org/10.1080/19312458.2021.1968361>
- Guess, A. M., Malhotra, N., Pan, J., Barberá, P., Allcott, H., Brown, T., Crespo-Tenorio, A., Dimmery, D., Freelon, D., Gentzkow, M., González-Bailón, S., Kennedy, E., Kim, Y. M., Lazer, D., Moehler, D., Nyhan, B., Rivera, C. V., Settle, J., Thomas, D. R., ... Tucker, J. A. (2023). How do social media feed algorithms affect attitudes and behavior in an election campaign? *Science*, 381(6656), 398–404. <https://doi.org/10.1126/science.abp93>
- Lorenzen, M. S. (2024). Appendixs til rapporten: Danskernes forståelse af, holdninger til og brug af (generativ) kunstig intelligens. University of Southern Denmark. <https://doi.org/10.21996/YNQA-RR56>
- Lorenzen, M. S., Vreese, C. de, & van Dalen, A. (2023). Danskernes digitale forståelse, digitale kompetecer og digitale adfærd—Og deres tillid til algoritmer. Syddansk Universitet. <https://doi.org/10.21996/7K7F-E379>
- Lorenzen, M. S. (2024). Danskernes forståelse af, holdninger til og brug af (generativ) kunstig intelligens. University of Southern Denmark. <https://doi.org/10.21996/YNQA-RR56>

References

- Malitesta, D., Cornacchia, G., Pomo, C., Merra, F. A., Di Noia, T., & Di Sciascio, E. (2025). Formalizing multimedia recommendation through multimodal deep learning. *ACM Transactions on Recommender Systems*, 3(3), 1–33. <https://doi.org/10.1145/3662738>
- Oeldorf-Hirsch, A., & Neubaum, G. (2025). What do we know about algorithmic literacy? The status quo and a research agenda for a growing field. *New Media & Society*, 27(2), 681–701. <https://doi.org/10.1177/14614448231182662>
- Smith, A. (2018, November 16). Attitudes toward algorithmic decision-making. Pew Research Center: Internet, Science & Tech. <https://www.pewresearch.org/internet/2018/11/16/attitudes-toward-algorithmic-decision-making/>
- Starke, C., Metikoš, L., Helberger, N., & de Vreese, C. (2025). Contesting personalized recommender systems: A cross-country analysis of user preferences. *Information, Communication & Society*, 28(1), 41–60. <https://doi.org/10.1080/1369118X.2024.2363926>
- You, S., Yang, C. L., & Li, X. (2022). Algorithmic versus Human Advice: Does Presenting Prediction Performance Matter for Algorithm Appreciation? *Journal of Management Information Systems*, 39(2), 336–365. <https://doi.org/10.1080/07421222.2022.2063553>